Experiment No. 7: Basic Video Processing Operations

Aim: To perform basic video processing operations using Python by reading a captured video and displaying it in normal speed, slow motion, and fast motion.

Program:

```
from google.colab import files

import cv2

from IPython.display import HTML, display

from base64 import b64encode


# Upload video

uploaded = files.upload()

video_path = list(uploaded.keys())[0]


def process_video(path, speed, output):

    cap = cv2.VideoCapture(path)


    fps = cap.get(cv2.CAP_PROP_FPS)

    width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))

    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))


    out = cv2.VideoWriter(

        output,

        cv2.VideoWriter_fourcc(*'mp4v'),

        fps * speed,

        (width, height)

    )
```

```
    while True:
        ret, frame = cap.read()
        if not ret:
            break
        out.write(frame)

    cap.release()
    out.release()

    with open(output, 'rb') as f:
        data = b64encode(f.read()).decode()

    display(HTML(f"""
    <h3>{output}</h3>
    <video width="500" controls>
        <source src="data:video/mp4; base64,{data}"
        type="video/mp4">
    </video>
    """))

# Normal speed
process_video(video_path, 1.0, "normal.mp4")

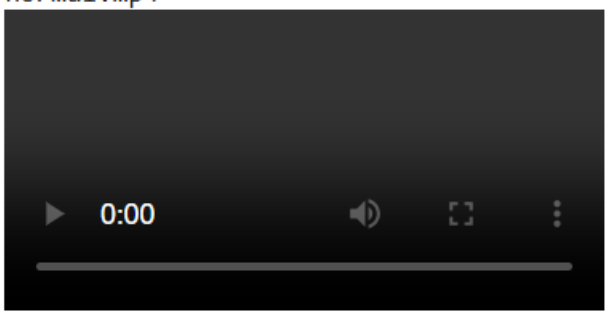
# Slow motion
process_video(video_path, 0.5, "slow_motion.mp4")
```

Fast motion

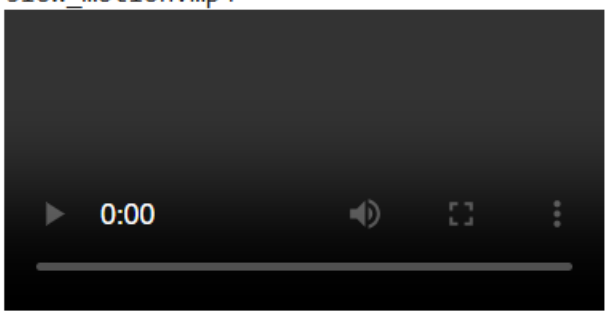
process_video(video_path, 2.0, "fast_motion.mp4")

OUTPUT:

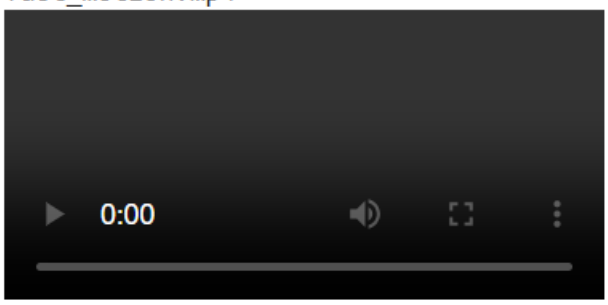
normal.mp4



slow_motion.mp4



fast_motion.mp4



Terminal

Expected Output:

1. Original video displayed at normal speed.
2. Video displayed in slow motion (0.5× speed).
3. Video displayed in fast motion (2× speed).

Result: The captured video was successfully read and processed using Python and OpenCV in Google Colab. The video was displayed at normal speed, slow motion, and fast motion.